

Unit 2, Lesson 4: Square Roots and Cube Roots

Benchmarks

MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.

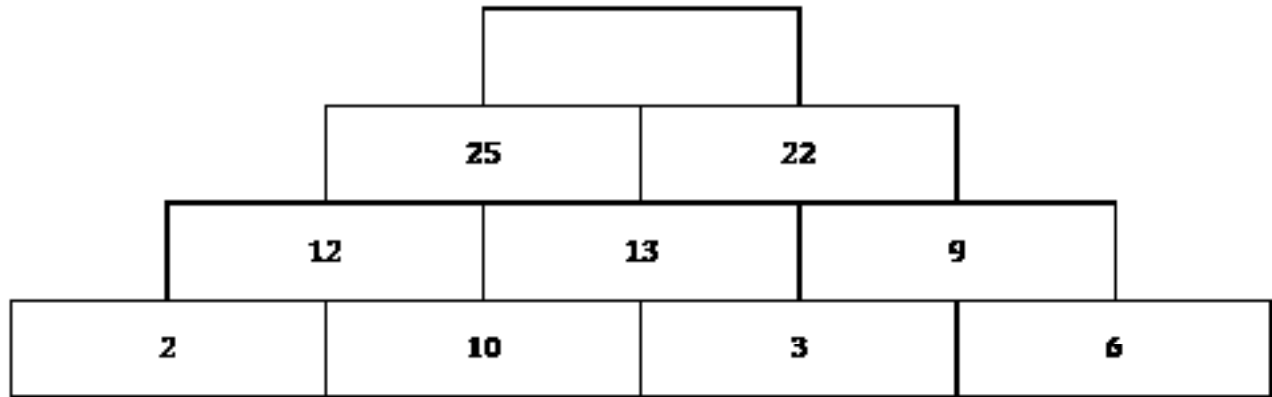
MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.

Learning Targets

- ☐ I can extend my understanding of square roots and cube roots.
- ☐ I can compare square roots and cube roots.
- ☐ I can approximate square roots and cube roots.

2.4.1 Warm-Up: Number Pattern

A pattern is shown with one value missing.



1. Determine the value of the empty cell in the pattern.
2. Explain how you determined the missing value.

2.4.2 Collaboration: Square Roots vs. Cube Roots

Work with your partner to complete the following.

1. Damion knows that $8^2 = 64$ and $4^3 = 64$. He says this means the following equations are all true.

Equation A	$\sqrt{64} = 8$
Equation B	$\sqrt[3]{64} = 4$
Equation C	$\sqrt{-64} = -8$
Equation D	$\sqrt[3]{-64} = -4$

- a. Discuss with your partner whether or not you agree that all of the equations Damion wrote are true.
- b. If Damion wrote any false equation(s), explain what makes them false. If all of Damion's equations are true, explain how you know.

2. Complete each statement.

- a. $\sqrt[3]{-1}$ is
☐ equal to _____
☐ not a real number
 because

- ☐ _____ $\times$ _____ $\times$ _____ $= -1$.
☐ _____ $\times$ _____ $= -1$.
☐ no real number multiplied by itself results in -1 .

- b. $\sqrt{-1}$ is
☐ equal to _____
☐ not a real number
 because

- ☐ _____ $\times$ _____ $\times$ _____ $= -1$.
☐ _____ $\times$ _____ $= -1$.
☐ no real number multiplied by itself results in -1 .

3. Work with your partner to complete the table by choosing a value of k based on the inequality. Then, determine if the result is real or non-real.

Description	Example	Real or Non-Real Result(s)
$\sqrt{k}$ where $k > 0$		☐ real ☐ non-real
$\sqrt{k}$ where $k < 0$		☐ real ☐ non-real
$\sqrt[3]{k}$ where $k > 0$		☐ real ☐ non-real
$\sqrt[3]{k}$ where $k < 0$		☐ real ☐ non-real

4. Write a short explanation of when a non-real result will occur.

2.4.3 Exploration: Square Roots of Non-Perfect Squares

There are an infinite number of perfect squares in the real number system.

1. Complete the list of the first 15 perfect squares.

$1^2 = \underline{\quad}$

$2^2 = \underline{\quad}$

$3^2 = \underline{\quad}$

$4^2 = \underline{\quad}$

$5^2 = \underline{\quad}$

$6^2 = \underline{\quad}$

$7^2 = \underline{\quad}$

$8^2 = \underline{\quad}$

$9^2 = \underline{\quad}$

$10^2 = \underline{\quad}$

$11^2 = \underline{\quad}$

$12^2 = \underline{\quad}$

$13^2 = \underline{\quad}$

$14^2 = \underline{\quad}$

$15^2 = \underline{\quad}$

2. Check your answers with your partner.

The square roots of perfect squares are rational numbers because they can be expressed as a ratio of two integers.

Square roots of non-perfect squares, such as $\sqrt{14}$, are *irrational numbers*.



Irrational numbers are real numbers that cannot be expressed as a ratio of two integers.

The decimal expansions of irrational numbers do not terminate nor repeat. Therefore, the values of a square root of a non-perfect square can only be approximated.



The ***principal square root*** is the positive square root of a positive real number.

When evaluating the square root of a number, the ***principal square root*** is assumed when the radical symbol is used. A negative symbol, $-$, is placed in front of the radical to indicate the negative square root. Perfect squares can be used to estimate the value of the principal square root of non-perfect squares.

3. Work with your partner to complete the following to approximate the principal square root of $\sqrt{14}$.
- a. Complete the inequality to show which two perfect squares 14 lies between.

$$\underline{\hspace{2cm}} < 14 < \underline{\hspace{2cm}}$$

- b. Use your answers to Part A to indicate which two square roots $\sqrt{14}$ lies between.

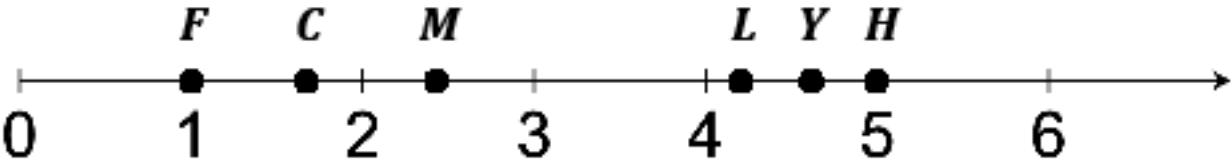
$$\underline{\hspace{2cm}} < \sqrt{14} < \underline{\hspace{2cm}}$$

- c. Discuss with your partner which square root the value of $\sqrt{14}$ is closer to.

- d. Plot $\sqrt{14}$ on the number line to indicate its approximate value based on your discussion.



The approximate locations of six square roots are plotted on the number line shown.



4. Work with your partner to determine which point matches the approximate value of each square root given in the table.

Square Root	Point
$\sqrt{3}$	
$\sqrt{21}$	
$\sqrt{1}$	
$\sqrt{6}$	
$\sqrt{18}$	
$\sqrt{25}$	

5. Complete the statement to explain how you determined the approximate location of the point selected.

Point

☐ C
☐ M
☐ L
☐ Y

 is between

☐ 1
☐ 2
☐ 3
☐ 4
☐ 5

 and

☐ 1,
☐ 2,
☐ 3,
☐ 4,
☐ 5,

 but

closer to

☐ 1
☐ 2
☐ 3
☐ 4
☐ 5

 because

☐ $\sqrt{3}$
☐ $\sqrt{21}$
☐ $\sqrt{6}$
☐ $\sqrt{18}$

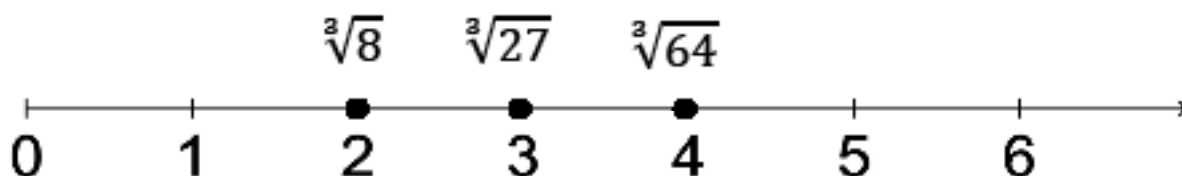
 is between $\sqrt{\quad}$ and $\sqrt{\quad}$, but closer to $\sqrt{\quad}$.

2.4.4 Exploration: Estimating Cube Roots

Similar to the square roots of non-perfect squares, the cube roots of some numbers are irrational.

The values of such cube roots can be approximated using the cube roots of perfect cubes.

1. A number line is shown with values indicated at specific points.



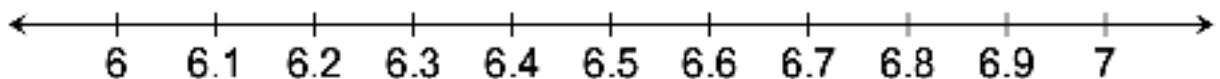
- a. Explain why $\sqrt[3]{8}$ is placed in the same location as 2 on the number line.
- b. Label the locations of the following cube roots on the number line.
 $\sqrt[3]{1}$, $\sqrt[3]{125}$, and $\sqrt[3]{0}$
- c. Plot and label the approximate locations of the following cube roots on the number line.
 $\sqrt[3]{9}$, $\sqrt[3]{10}$, and $\sqrt[3]{140}$
- d. Compare your answers to your partner's. Discuss any differences and make revisions to your work as needed.

2.4.5 Guided Practice: Locating Square Roots and Cube Roots on a Number Line

1. Consider $\sqrt{42}$.

- a. Between which two perfect squares does 42 lie?
- b. Between which two integers is the value of $\sqrt{42}$?
- c. Corey said, "42 is about halfway between 36 and 49, but slightly closer to 36, so I think a good estimate for $\sqrt{42}$ would be about halfway between 6 and 7, but slightly closer to 6."

Circle two of the values shown on the number line that Corey could use to check his thinking.



- d. Square both values circled in Part C to determine which is a better approximation for $\sqrt{42}$.
- e. Plot and label the approximate location of $\sqrt{42}$ on the number line above based on your work in Part D.

2. Six equations are shown.

$$a = \sqrt{9}$$

$$b^2 = 8$$

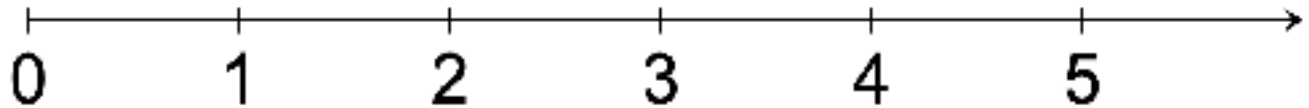
$$\sqrt{10} = q$$

$$1 = n^2$$

$$k = \sqrt[3]{24}$$

$$\sqrt[3]{64} = y$$

- a.** Plot and label the approximate locations of a, b, q, n, k , and y on the number line.



- b.** Choose one value and explain how you determined its location on the number line.

2.4.6 Wrap-Up: Growth Mindset

**"Don't be afraid of failure. This is the way to succeed."
– LeBron James**

- 1. Complete each statement.**
 - a. Today I found a way to succeed at...**

 - b. Today I did not give up on learning...**